

Computational Modeling And data entry

```
GTGACAAAAGAAAAAAAAATCCTTTTCATTACACATG
CTTTCTGATGCAACGGGGGAAACATTAATCTCTGTT
GGAAGAGCTGTCGCATCCCAGTATACAACGTATCAG
GCAACAGAGCATATCTATCCTATGATTTCGTAATAAAAT
ACAGTTAGAAAAAGCTCTTGATGAAATAAAACAAG
AGCCAGGTATTGTTCTTTACACAATGATTGATGAAG
ATCTAGTAACTATTTTGAAGCAGATGTTATTCTTGTA
GGAATCTCAAGAACCTCTAAAACGCCAACAGATCCA
GTCAAAGCGTTTAGATCCTAGCAATCGGATTCAGTT
```

DNA Sequence

In-silico digestion (XbaI)

DNA fragment length (Kbp)
(Error equals Zero)

Fragment length (Kbp)	Repetition
100	1
80	1
60	1
60	1

Data entry

Storage

comparisons

SQL database Server

Internet

Analysis

Repetition	Fragment accumulation factor
1	2
1	1
1	3

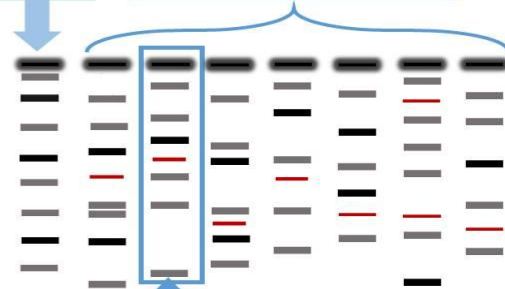
Results back to the client

Bacterial strain typing using DNA Pulsed-field gel electrophoresis

Results

Marker

Bacterial DNA digests



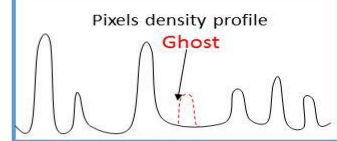
PFGE image

Calculations

Marker equation

Intensity equation

DNA fragment length (Kbp)
(Error NOT equals Zero)



Fragment length (Kbp)	Fragment accumulation factor
100	$\geq 1 \rightarrow (2)$
70	$\geq 1 \rightarrow (1)$
60	$\geq 1 \rightarrow (3)$

